

The University of Nottingham
Malaysia Campus

SCHOOL OF COMPUTER SCIENCE

A LEVEL 3 MODULE, SPRING SEMESTER 2023-2024

**COMPUTER SECURITY
(COMP3028)**

Time allowed ONE Hour

Candidates must NOT start writing their answers until told to do so

Answer ALL THREE questions

Only silent, self-contained calculators with a single-line display are permitted in this examination.

Dictionaries are not allowed with one exception. Those whose first language is not English may use a standard translation dictionary to translate between that language and English provided that neither language is the subject of this examination. Subject specific translation dictionaries are not permitted.

No electronic devices capable of storing and retrieving text, including electronic dictionaries, may be used.

DO NOT turn examination paper over until instructed to do so

INFORMATION FOR INVIGILATORS: Please ensure that the Exam Paper is also collected at the end of the examination.

1. The following questions are about computer security in general.
 - a) Three aspects of security are prevention, detection and reaction. Write a paragraph explaining why methods used for the prevention of, detection of and reaction to, theft of physical property, may not be appropriate when the crime involves the theft of digital information.
(6 marks)
 - b) A password system is used to provide authentication and access control.
 - i) Define the terms authentication and access control.
(2 marks)
 - ii) What does it mean to say that an attacker performs an intelligent search, a dictionary search, and an exhaustive search for a user's password?
(3 marks)
 - c) Please classify each of the following as a violation of confidentiality, integrity, availability:
 - i) A copies B's homework.
 - ii) A crashes B's operating system.
 - iii) A changes the amount on B's check from 100 to 1000.
(3 marks)
 - d) List and explain *three* network threats that a firewall does not protect against. (If a threat only applies to certain types of firewalls, then explain why this is the case.)
(6 marks)

2. The following questions are about cryptography.

- a) State *two* reasons why modern ciphers, whether secret-key or public-key, perform encryption on blocks of data, as opposed to encrypting single bits or single letters.
(4 marks)
- b) State the two main “hard” problems used as the basis for public-key encryption schemes, and explain why each is believed to be secure.
(4 marks)
- c) Explain how the RSA public-key cryptosystem works. Your answers should include the generation of public and private keys, the encryption equation, the decryption equation, and the basis for the security of RSA.
(12 marks)

3. The following questions are about access control.

- a) In a University department, there is an administrator (Mrs A), three lecturers (Dr B, Dr C and Dr E)) and a Head of Department (Prof H). Three documents concerning the department are *salary.doc*, *timetable.doc* and *marks.doc*.

The Head of Department has permission to read all of the documents and has write access to *salary.doc*. The administrator has read and write access to *timetable.doc* and read access to *salary.doc*. The lecturers each have read access to *timetable.doc* and write and read access to *marks.doc*.

Information regarding the permissions each staff member has regarding the documents is to be stored in an access control table indexed by subjects and objects.

- i) List the subjects, objects and operations, which will be used in the access control table. (3 marks)
- ii) Represent the information given as an access control table. (5 marks)
- b) Contrast the following two types of access control mechanisms, Access Control List and Capability, in terms of (i) ease of making an authorisation decision during execution; (ii) ease of adding access for a new subject; (iii) ease of deleting access by a subject; and (iv) ease of creating a new object to which all subjects by default have access to. Justify your answers to the questions. (12 marks)